

Enhanced fluid mixing using a novel multi-stage static pipeline mixer: Numerical simulation and mechanism analysis

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ABSTRACT

Static pipeline mixers play a crucial role in water treatment processes, and their mixing efficiency can be influenced by factors. Innovating mixer design to simultaneously reduce head loss and G-values while maintaining effective mixing across various flow conditions holds practical significance for improving treatment process performance. This study used a combination of numerical simulations and experimental validation to investigate the mixing process of a novel multi-stage static pipeline mixer. The results indicated that within the flow velocity range of 0.2–1.8 m/s, compared to the Kenics mixer, the new multi-stage static pipeline mixers (HOCM and COCM) achieved significant reductions in head loss (89.4 % for HOCM and 52.9 % for COCM) and G-values (67.3 % for HOCM and 31.3 % for COCM). The turbulent kinetic energy intensity of HOCM was much lower than that of COCM and Kenics mixers. Importantly, all mixers reached a state of complete mixing with the coefficient of variation (CoV) less than 0.05. In the HOCM mixer, the 2nd mixing stage was identified as the primary mixing zone, while in the COCM mixer, the 1st mixing stage played a crucial role in mixing. When the mixers achieved a mixing degree of 95 %, the head loss of HOCM and COCM was significantly lower than various types of traditional mixers (Kenics, SMX, SMV, ISG, Inliner Lightnin). This is attributed to the highly efficient mixing effect generated by the clockwise helical flow produced in the 1st and 2nd mixing stages.

1. Introduction

Static pipeline mixers are the most commonly used mixing facilities, with small installation footprint, low maintenance costs, and short residence times, which can be widely applied in the mixing processes of numerous fields, such as polymer processing [1], biopharmaceuticals [2], water treatment [3,4], food processing [5], and petrochemicals [6]. In the field of water treatment, multi-stage static pipeline mixers are commonly used facilities for coagulant dosing and mixing. The multi-stage static pipeline mixer consists of multiple sets of fixed mixing units, offering advantages such as minimal space occupation, high reliability, continuous fast mixing, and low maintenance costs [7]. The mixing process is powered by the kinetic energy of water flow, and relies on the hydraulic motion generated as water flows through the mixing units to facilitate the uniform mixing of coagulants with water [8]. The structural design of mixing units is typically quite complex, which in turn leads to practical issues with traditional mixers (such as Kenics mixers), including significant head loss, high G-values, susceptibility to

variations in water flow affecting mixing effectiveness, and limited applicability to different operating conditions [9]. The number of mixing blades [10], the twist angle, and the length-to-diameter ratio (L/D) [11] are parameters that significantly influence the mixing process of Kenics mixers. Efforts to optimize mixer blade structures and introduce surface perforations have proven ineffective in reducing head loss and energy consumption, thereby limiting the development and application of traditional water treatment mixers [12]. In recent years, various novel multi-stage static pipeline mixers with simple and unique mixing structures have been proven effective in reducing mixing resistance and improving mixing efficiency [13–15]. However, these new mixers vary significantly in size and mixing conditions [16–18], and there is a scarcity of research reports on their application in water treatment mixing processes [19].

In water treatment processes, the mixing time is short and the velocity fluctuations are significant [20], and the fluid dynamics are complex [21]. The mixing efficiency is greatly influenced by the type of mixer [22], making it difficult to intuitively express the local details of

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the mixing process [23]. Direct characterization methods for mixing efficiency are limited, often relying on indirect indicators such as subsequent coagulation and sedimentation effects to reflect mixing efficiency. Qualitative methods for characterizing mixing effects mainly include color dye dilution [22], fluorescence substance dilution [24], and colorant generation techniques [25]. These methods require mixers to have observable external chamber structures to observe the vortex structures and mixing states perpendicular to the flow direction [26].

The aforementioned characterization methods cannot accurately represent the mixing differences among various mixing units. Computational fluid dynamics (CFD) numerical simulation techniques can systematically characterize the mixing patterns of each mixing unit, thereby overcoming the limitations of experimental characterization methods. This approach assists in analyzing mixing mechanisms from a microscopic fluid dynamics perspective [27,28]. CFD numerical simulations commonly utilize three main models: direct numerical simulation (DNS) [29], large eddy simulation (LES) [30], and reynolds-averaged navier-stokes (RANS) [31]. CFD numerical simulations mixers such as Kenics [30], SMX [32], corrugated [33], and T-type [29] have consistently demonstrated accurate predictive capabilities, clearly revealing local vortex structures with high turbulent kinetic energy and changes in the flow field during the mixing process [27].

This study aimed to address the common issues associated with traditional static pipeline mixers, such as significant head loss, high G-values, and susceptibility to variations in flow rate. By utilizing a multiphase flow mixing model based on the RANS model and Fluent software, we innovatively developed and investigated a novel multi-stage static pipeline mixer for the mixing process [34]. First, the mixing process of the 3D printed novel multi-stage static pipeline mixer was compared under simulated and actual scaled-down experimental setups to determine the accuracy of the mathematical model used. Then, the main parameters of the mixing process were compared between the actual sizes of Kenics and the novel mixer, and the mixing mechanism of the novel mixer was analyzed. The mixing efficiency of the novel mixer was compared with traditional mixers (Kenics, SMX, SMV, ISG, Inliner Lightning), and the significant advantages of the new mixer were evaluated. The result provided the theoretical basis for the assessment of practical mixing performance.

2. Materials and methods

2.1. Physical model of multi-stage static mixer

The model of each mixer is established using SolidWorks software, and the structure is rendered using Cinema 4D (Fig. 1). The traditional Kenics mixer, hollow outer-cavity multi-stage static pipeline mixer (referred to as HOCM) and the closed outer-cavity multi-stage static pipeline mixer (referred to as COCM) all have the length of 2000 mm, the effective mixing length of 1000 mm, and the inner diameter of the inlet pipe is 200 mm (Fig. 1a). The Kenics mixer has a dosing pipe with an inner diameter of 40 mm and is equipped with 5 helical mixing blades, each of which has a ratio length-to-diameter of 1 and a twist angle of 180° . The blades are arranged in an alternating pattern of left-hand and right-hand twist units, and each left-hand and right-hand unit is staggered by 90° within the pipeline. The dosing for HOCM and COCM is carried out using a ring-shaped pipe, with a straight inner diameter of 40 mm and a ring inner diameter of 20 mm.

The mixers feature a double-cavity structure, with eight hollow concave spiral diversion troughs in the inner cavity, which are twisted at an angle of 360° (Fig. 1b). A portion of the inflow from the inner cavity is guided by the diversion trough to the outer cavity, mixed with the coagulant solution in the first stage, and then flows back to the inner cavity through the collection groove at the rear of the mixer, whose layout is symmetrically mirrored that of the diversion trough. The HOCM allows the outer-cavity water flow to pass through both the diversion trough and the gaps in the outer-cavity, while the COCM confines the outer-cavity water flow to the diversion trough. The ring-shaped dosing pipe section is equipped with eight dosing holes with an inner diameter of 10 mm, corresponding to the diversion trough. The 3D printed mixers (3D-HOCM and 3D-COCM) with equal scale reduction of 10 times are used as the experimental setup for the multi-stage static pipeline mixer (Fig. 1c). The total length of the mixing pipe of the 3D-HOCM and 3D-COCM is 200 mm and the diameter of inlet pipe is 20 mm. The inner diameter of straight section for dosing is 4 mm, the inner diameter of ring section is 2 mm, and the inner diameter of outlet hole is 1 mm.



Fig. 1. Physical models of Kenics, HOCM, and COCM static mixers (a), flow and dosing components of static mixers with diversion trough, collection groove, and ring-shaped pipe mixers (b), and physical models of 3D-HOCM and 3D-COCM (c).

2.2. Mathematical model and simulation method

2.2.1. Governing equations

Assuming that the mixing process meets the conditions of constant fluid, incompressible, three-dimensional turbulent flow and constant temperature [35], the multi-phase flow Mixture Model is used as the mathematical model to solve the mass conservation, momentum conservation, and turbulence model governing equations of continuous flow. The mass conservation equation is as follows:

$$\frac{\partial(\rho\nu_i)}{\partial x_i} = 0 \quad (1)$$

where ρ stands for density in kg/m^3 , ν_i stands for the velocity vector.

Momentum conservation equation:

$$\frac{\partial}{\partial x_j}(\rho\nu_i\nu_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial \nu_i}{\partial x_j} + \frac{\partial \nu_j}{\partial x_i} \right) \right] \quad (2)$$

where ρ stands for density in kg/m^3 , μ represents the dynamic viscosity of the fluid in $\text{kg/(m}\cdot\text{s)}$.

Turbulence model governing equation is composed of turbulent kinetic energy equation and turbulent kinetic energy dissipation rate equation.

Turbulent kinetic energy equation:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k \nu_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{\text{eff}} \frac{\partial k}{\partial x_j} \right) + G_k + \rho \epsilon \quad (3)$$

Turbulent kinetic energy dissipation rate equation:

$$\frac{\partial(\rho \epsilon)}{\partial t} + \frac{\partial(\rho \epsilon \nu_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{\text{eff}} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon}^* \frac{\epsilon}{k} G_k - C_{2\epsilon} \rho \frac{\epsilon^2}{k} \quad (4)$$

where $\mu_{\text{eff}} = \mu + \mu_t$, $\mu_t = \rho C_\mu \frac{k^2}{\epsilon}$, $C_{1\epsilon}^* = C_{1\epsilon} - \frac{\xi(1-\xi/\xi_0)}{1+\beta\xi^3}$, $\xi = (2E_{ij} \cdot E_{ij})^{1/2} \frac{k}{\epsilon}$, $E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$.

2.2.2. Numerical method and boundary condition setting

The transient three-dimensional spatial model is used for numerical simulation, with a gravitational acceleration of -9.81 m/s^2 . The Standard k- ϵ model is used for turbulence model governing equation. The SIMPLE algorithm is adopted to carry out the Velocity-pressure coupling [36]. The Standard Method is used for the discrete format of pressure terms, and the first-order upwind is used for the discrete format of other terms, ensuring rapid convergence in the calculation process [12]. Residual accuracy is controlled within 10^{-6} .

Boundary conditions are set with water as the primary phase, having a density of 998.2 kg/m^3 and dynamic viscosity of $0.0010 \text{ kg/(m}\cdot\text{s)}$. The coagulant solution is considered the secondary phase, with a density of 1000.0 kg/m^3 and dynamic viscosity of $0.0011 \text{ kg/(m}\cdot\text{s)}$. The inlet boundary conditions of the mixer are defined as velocity-inlet, and the outlet boundary conditions are set to outflow. The pipe walls are no-slip walls by default.

In the simulation process, the inlet flow velocity of the primary and secondary phases in the mixer are equal. The inlet flow velocity ranges from 0.2 to 1.8 m/s and 0.2 to 0.8 m/s in multi-stage static pipeline mixer with actual size and in 3D printing mixers, respectively. The Reynolds number of the primary phase ranges from $40,000$ to $360,000$ and 4000 to $16,000$ in multi-stage static pipeline mixer with actual size and in 3D printing mixers, respectively. All within the turbulent state. Reynolds number is calculated using the following equation:

$$Re = \frac{\rho u d}{\mu} \quad (5)$$

where ρ stands for density in kg/m^3 , u is the velocity in m/s , d is the pipe diameter in meters (m), and μ is the dynamic viscosity in $\text{kg/(m}\cdot\text{s)}$.

2.3. Experimental method

2.3.1. Experimental setup

The experiments are carried out to determine the accuracy of the model used in the study, the details of which are provided in Fig. S1 and Text S1. The inlet flow velocity ranges from 0.2 to 0.8 m/s in experiments. The inner cavities and dosing pipes of 3D-HOCM and 3D-COCM (SLA process, WeNext, Shenzhen WeNext Technology Co., Ltd., China) are made of white resin, and the outer cavities are made of transparent photosensitive resin.

2.3.2. Head loss and G-value

In experimental processes, water head loss can be directly measured and calculated using pressure gauges that are set up (usually $1 \text{ m H}_2\text{O} = 10,000 \text{ Pa}$). The formula for calculating head loss in a simulation process is as follows:

$$h = h_{2,\text{mean}} - h_{1,\text{mean}} \quad (6)$$

where h is the head loss in the mixer, in meters (m); $h_{2,\text{mean}}$ is the mean pressure at various control points in the outlet section, in meters (m); $h_{1,\text{mean}}$ is the mean pressure at various control points in the inlet section, in meters (m).

The mixing intensity of the mixing process can be expressed using the G-value, calculated as follows:

$$G = (gh/\nu T)^{1/2} \quad (7)$$

h and T can be represented as:

$$h = \zeta \cdot u^2 / 2g \quad (8)$$

$$T = L/u \quad (9)$$

The relationship between G-value and u can be derived from formula (7) to (9):

$$G = (\zeta/2\nu L)^{1/2} \cdot u^{3/2} \quad (10)$$

where g is the gravitational acceleration, 9.8 m/s^2 ; h is the head loss in the mixer, in meters (m); ν is the kinematic viscosity of water, in m^2/s ; T is the hydraulic retention time, in seconds (s); ζ is the local resistance coefficient; u is the inlet flow velocity, in m/s ; L is the axial length of the mixer, in meters (m).

For a specific mixer, $(\zeta/2\nu L)^{1/2}$ can be considered a constant. Therefore, the G-value and the inlet flow velocity u have a power-law relationship, with an exponent of $3/2$. This means that the G-value is directly proportional to the inlet flow velocity u raised to the power of $3/2$. This relationship signifies that the mixing intensity, as represented by the G-value, increases with the cube root of the inlet flow velocity.

2.3.3. Qualitative and quantitative characterization of mixing effectiveness

The variation of local turbulent kinetic energy during the mixing process can reflect the distribution of vortices, with regions of high turbulent kinetic energy typically associated with strong hydraulic vortices, representing the primary areas of mixing. The distribution of turbulent kinetic energy within the mixer during numerical simulations can be directly obtained through software simulation. The calculation formula is as follows:

$$\text{Turbulent kinetic energy (TKE)} = \frac{1}{2} \rho \overline{u^2} \quad (11)$$

where ρ is the fluid density, in kg/m^3 ; u is the local instantaneous flow velocity, in m/s .

The mixing effectiveness is qualitatively analyzed by dilution of colored dyes method. Water-soluble rose-red dye is used to simulate the coagulant, and the mixing process is observed frame by frame by shooting videos with a single lens reflex camera (Canon 5D Mark IV,

video FHD 50p format).

The absorbance detection method is used for quantitative analysis [37]. The absorbance of the water-soluble rose-red dye is measured at a wavelength of 542 nm, and a standard curve of absorbance values is established (Fig. S2). Ten sampling tubes are simultaneously inserted at the outlet of the mixing pipeline to sample and detect the absorbance values, and the coefficient of variation (CoV) of the coagulant solution concentration distribution at the outlet cross-section is calculated [38]. The CoV values in both numerical simulation and experimental processes are calculated as follows:

$$CoV = \sqrt{\frac{\sum_{j=1}^N (\Phi_{S,j} - \Phi_{S,mean})^2}{N-1}} \cdot \frac{1}{\Phi_{S,mean}} \quad (12)$$

where $\Phi_{S,j}$ is the volume fraction of the coagulant solution at the j th computational section of each control volume, $\Phi_{S,mean}$ is the average volume fraction of the coagulant solution in that section, and N is the total number of samples in that section.

2.3.4. Calculation of head loss in various types of traditional mixers

The head loss in traditional static pipeline mixers is typically represented using the Darcy-Weisbach equation:

$$\Delta P = \lambda \frac{L}{D} \rho \frac{u^2}{2} \quad (13)$$

where λ is the local friction factor, L is the length of the mixing stage, in meters (m); D is the diameter of the mixing pipe, in meters (m); ρ is the fluid density, in kg/m^3 ; and u is the flow velocity, in m/s.

3. Results and discussion

3.1. Simulation and validation of the mixing process in 3D printing mixers

3.1.1. Pressure variation trend

To validate the accuracy of the mathematical model, a comparison

between simulations and experiments was conducted using the experimental setup of the multi-stage static pipeline mixer (3D-HOCM and 3D-COCM). The variation of simulated and experimental head losses for each type of mixer is shown in Fig. 2a. The head losses and flow velocities exhibit a quadratic relationship, which aligns with the fundamental relationship of water head losses and flow velocities described by Eq. (8). Within the flow rate range of 0.2 m/s to 0.8 m/s, the simulated values of local head losses for both mixer types closely match the experimental values. The mean relative error between experimental and simulated values is $\pm 4.86\%$ (3D-HOCM) and $\pm 2.61\%$ (3D-COCM), indicating that the Mixture Model can effectively predict the pressure changes in the actual mixing process. As the inflow velocity increases, the head losses exhibit a growing trend, reaching 0.66 m (3D-HOCM) and 1.38 m (3D-COCM) at a flow velocity of 0.8 m/s. The growth rate of head loss of the 3D-COCM is higher than that of 3D-HOCM, indicating that the local resistance of the closed outer-cavity is significantly higher than that of the hollow outer-cavity design.

The variation of G-values for each type of mixer is shown in Fig. 2b. The G-values under both simulated and experimental conditions follows a cubic root function relationship with flow velocity, which is in line with the derived formula (10). The G-values obtained from the measured head loss and the G-values calculated from Eqs. (6)–(9) for the simulation process are also quite close to each other, and the relative error between experimental and simulated values is approximately $\pm 2.52\%$ for 3D-HOCM and $\pm 1.33\%$ for 3D-COCM, respectively. Within the flow velocity range of 0.2–0.8 m/s, the simulated G-values for 3D-HOCM fall within the range of 666 s^{-1} to 5147 s^{-1} , while the simulated G-values for 3D-COCM fall within the range of 959 s^{-1} to 7310 s^{-1} . The G-values for 3D-COCM are consistently higher than those for 3D-HOCM under all flow velocity conditions, indicating that 3D-COCM exhibits a higher mixing intensity compared to 3D-HOCM.

3.1.2. The variation patterns of mixing efficiency

When the cross-sectional CoV value is less than 0.05, the mixture can be considered to be completely mixed, and in practice the mixing can be considered to have reached a uniform state when a value is less than 0.1

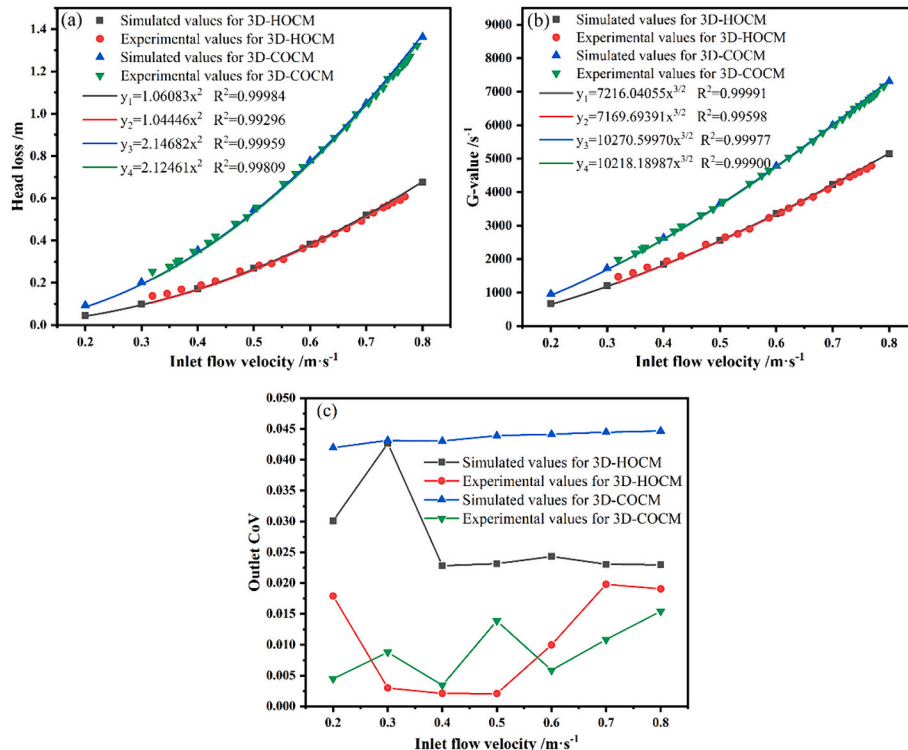


Fig. 2. Variations of parameters that head loss (a), G-value (b), and CoV (c) in the mixing process of 3D Printing mixers under simulations and experiments.

[36]. The experimental and simulated values of CoV are both lower than 0.05 (Fig. 2c), which indicates that the designs of the 3D-HOCM and 3D-COCM mixers can achieve rapid mixing of water and dye. By comparing the simulated values and the experimental values of different mixers, it is found that the experimental CoV values are consistently lower than the simulated values (Fig. 2c), which indicates that the mixing performance under experimental conditions is better than that of the simulated conditions, primarily due to differences in number of samples between the simulation and experiment [19,39].

The simulated CoV value of 3D-HOCM is consistently lower than that of 3D-COCM at different flow rates, however, the measured CoV value of 3D-HOCM at different flow rates is not much different from that of 3D-COCM at a low level below 0.020, which indicates that the mixing performance of the two mixers is the same in the practical mixing processes. The design of 3D-HOCM can reduce the head loss and G-values while ensuring the mixing performance, which results in lower pressure losses during the mixing process with higher economy. Therefore, 3D-HOCM can be considered as the preferred design option for mixers, in practical applications, it should ensure that the 2nd mixing stage has sufficient length to ensure the full development of the swirl process and achieve uniform mixing performance.

3.1.3. Patterns of flow variation

Comparing the simulation results based on the Mixture Model and the actual mixing process through direct observation provides a more intuitive representation of the development and changes in the internal flow field within the mixer (Fig. 3). When the coagulant solution volume fraction under CFD simulation conditions is 0.04, the solution concentration reaches its theoretical maximum. For 3D-HOCM, when the dye enters the 1st mixing stage, it diffuses throughout the entire outer-cavity flow channel and reaches the outlet cross-section after 2.00 s (Fig. 3a). Spiral flow is formed in both the 1st and 2nd mixing stages, indicating that the spiral guide grooves can induce a forced change in direction in the 1st mixing stage and affect the flow pattern in the 2nd mixing stage. The simulation results clearly illustrate the mixing details of 3D-HOCM that the dye first enters the outer cavity through the upper dosing hole and then rotates 360° along the spiral direction shown by the arrow after

0.25 s to enter the 2nd mixing stage (Fig. 3b). There are regions with lower coagulant solution volume fractions near the pipe wall of 3D-HOCM (highlighted in the green box) in the initial part of the 2nd mixing stage, however, the mixture reaches a relatively uniform state at the outlet cross-section (Fig. 3a, b).

For 3D-COCM, due to the closed outer-cavity design, the coagulant solution can only flow within the spiral guide grooves, resulting in the reduction of the volume of the outer-cavity in the 1st mixing stage and the hydraulic residence time within the mixer, and the dye diffuses to the outlet cross-section after 1.00 s (Fig. 3c). The recording process also indicates that the swirl formed in the 2nd stage enables the rapid dispersion of the dye from the 1st mixing stage to the entire straight pipe section of the 2nd mixing stage. The simulation results of 3D-COCM clearly show that the solution in the 1st mixing stage exhibits a right-handed helical flow pattern and rapidly achieves a uniform state in the 2nd mixing stage (Fig. 3d). There is no low-volume fraction region near the pipe wall in the 2nd mixing stage with 3D-HOCM, mainly because the G-value of the 3D-COCM is higher than that of the 3D-HOCM, resulting in a greater mixing intensity, which is more conducive to rapid mixing of the solution in the 2nd mixing stage [40].

3.2. Numerical simulation of multi-stage static pipeline mixer with actual size

The Mixture Model established by Eqs. (1)–(4) can effectively reflect the changes in head loss, G-value, mixing efficiency, and flow pattern for each mixer during the mixing process according to the simulation and experimental results of the mixing process in Section 3.1, which demonstrates that the model can be used to simulate the mixing process of full-scale multi-stage static pipeline mixers due to high accuracy in turbulent simulation processes. In the following research, the key parameter variations of traditional Kenics mixers and various multi-stage static pipeline mixers during the mixing process will be compared under numerical simulation, in order to better explain the mixing mechanisms.

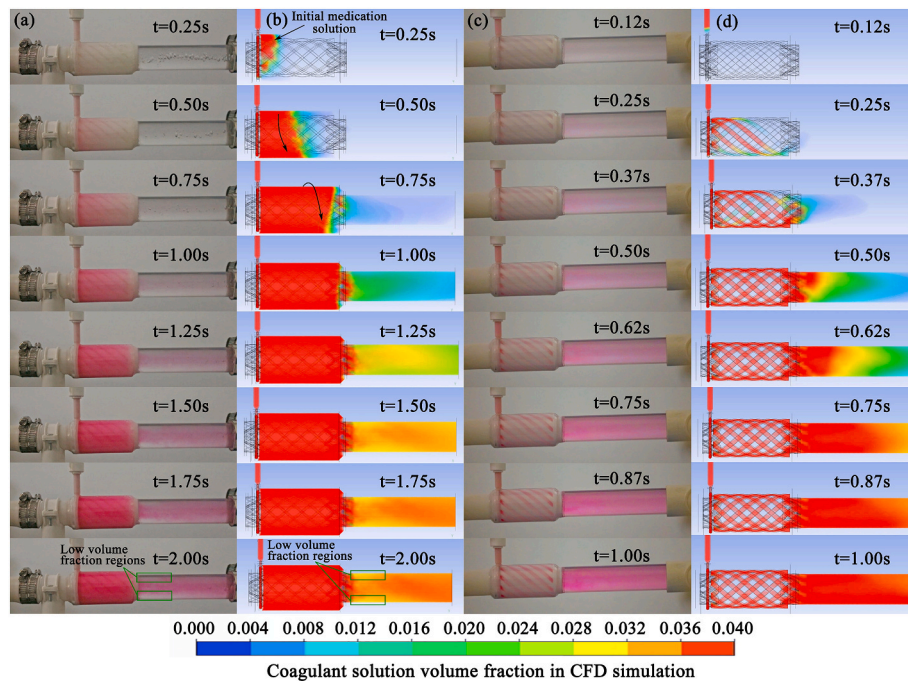


Fig. 3. The simulation results based on the Mixture Model and the actual mixing process of water-soluble rose red dye with water in 3D-HOCM (a and b, respectively) and 3D-COCM (c and d) at a flow rate of 0.5 m/s.

3.2.1. Pressure variation trend

The water head loss for each full-scale mixer, after nonlinear fitting, exhibits a quadratic relationship with the flow velocity (Fig. 4a), which also aligns with the fundamental relationship described by Eq. (8) for water head loss and flow velocity. With the increase of the inlet flow velocity, the head loss shows a growing trend, but the magnitude of the increase is different. The head loss of the Kenics mixer increase the fastest, followed by the COCM and the HOCM. It indicates that compared with the Kenics mixer, the design of HOCM and COCM mixer can significantly reduce the head loss during the mixing process, with an average reduction of 89.2 % (HOCM) and 52.6 % (COCM), respectively. The head loss of Kenics mixer, as a representative of low-resistance mixers, is already at a relatively low level compared to various traditional mixers and commercial mixers, e.g., Komax, SMX, LPD, and LLPD [36,41]. Ouda et al. further developed Gyroid, Diamond, Primitive, and IWP mixers based on Kenics mixers, which reduced energy consumption by more than 25 % compared with standard Kenics, produced lower head loss and achieved better mixing effect [42]. In this study, the novel multi-stage pipe mixer can significantly reduce the head losses during the mixing process (more than 50 % compared to the standard Kenics), therefore, the HOCM and COCM have great advantage in reducing the energy consumption in the mixing process.

The relationship between G-values and inlet flow velocity also conforms to the Eq. (10), which shows a well-fitted cubic root variation pattern (Fig. 4b). When the inlet flow velocity ranges from 0.2 m/s to 1.8 m/s, the G-value of the Kenics mixer is in the range of 225–6057 s⁻¹, and the G-value of the HOCM and COCM mixer is in the ranges of 81–1981 s⁻¹ and 160–4162 s⁻¹, respectively. The G-value of HOCM and COCM mixer is lower than that of the Kenics mixer, which decreases by 67.2 % and 31.1 %, respectively. In this study, the G-value of the HOCM mixer (at flow velocities of 0.2–1.8 m/s) and the COCM mixer (at flow velocities of 0.2–1.4 m/s) are lower than the optimal range (3000–5000 s⁻¹) [40]. It was not accurate to judge the mixing effect solely based on G-values, and some types of mixers can also achieve uniform mixing at low mixing intensity [43,44], especially for new types of mixers. Therefore, it remains necessary to further investigate the mixing performance of this type of mixer beyond just G-values.

3.2.2. The pattern of changes of turbulent kinetic energy

The distribution pattern of turbulent kinetic energy during the mixing process for each mixer is illustrated in Fig. 5. The turbulent kinetic energy distribution of the Kenics mixer (Fig. 5a) indicates the presence of high turbulent kinetic energy regions at the boundaries of each mixing stage. The turbulent kinetic energy can reach a maximum of

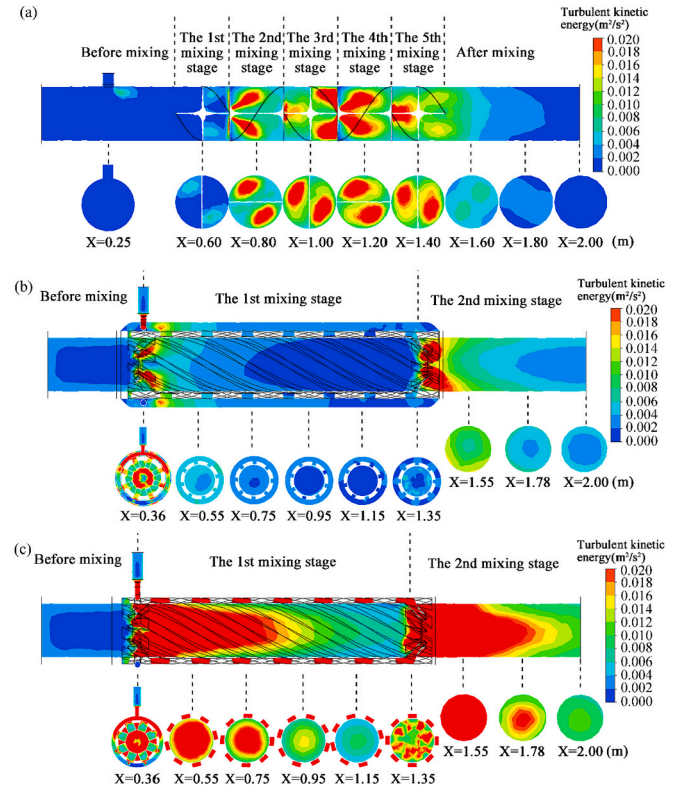


Fig. 5. The turbulent kinetic energy at different cross-sections of Kenics (a), HOCM (b) and COCM (c).

0.018–0.020 m²/s². Symmetrically, two high turbulent kinetic energy regions form on either side of the mixing blades at each stage. After mixing, the intensity of these two high turbulent kinetic energy regions gradually diminishes.

From the turbulent kinetic energy distribution of the HOCM (Fig. 5b), it can be observed that at distance of 0.36 m in the 1st mixing stage, the turbulent kinetic energy reaches its maximum, ranging from 0.018 to 0.020 m²/s². With an increase in the mixing distance of the 1st mixing stage, the turbulent kinetic energy gradually decreases. Within distance of 1.35–1.55 m in the 2nd mixing stage, the turbulent kinetic energy rapidly increases, reaching 0.014–0.020 m²/s². However, within distance of 1.55–2.00 m, the turbulent kinetic energy decreases,

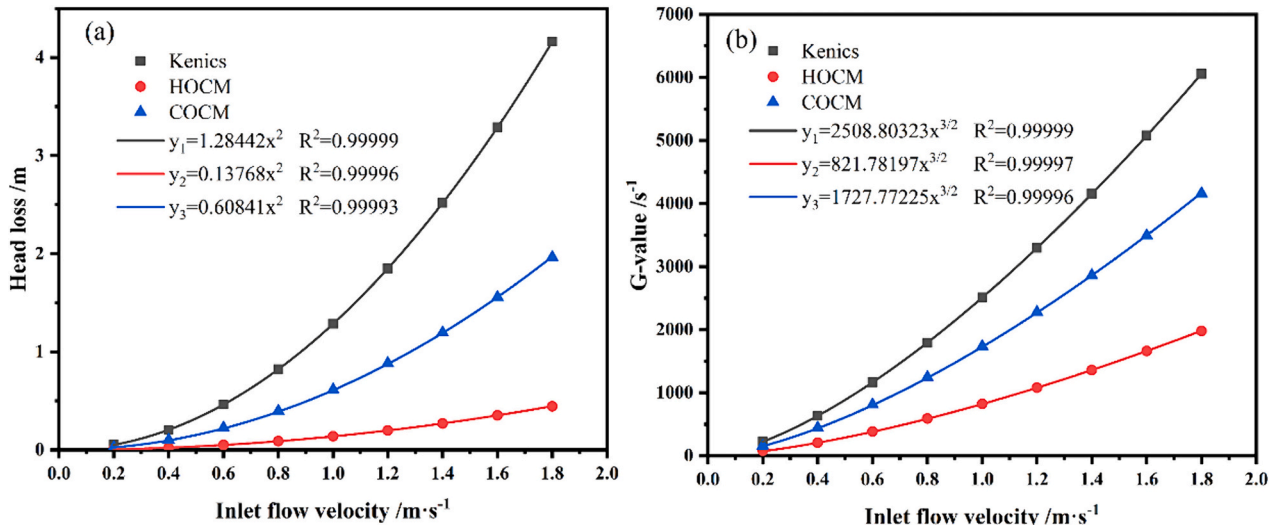


Fig. 4. Variations of parameters that head loss (a) and G-value (b) in the mixing process of actual sized mixers under numerical simulation.

exhibiting a distribution pattern where the turbulent kinetic energy at the periphery of the pipe is greater than that at the pipe's axis.

From the turbulent kinetic energy distribution of the COCM (Fig. 5c), it can be observed that within distance of 0.36–0.80 m in the 1st mixing stage, the turbulent kinetic energy inside the cavity reaches $0.016\text{--}0.020\text{ m}^2/\text{s}^2$, while in the spiral diversion troughs, the turbulent kinetic energy also reaches $0.018\text{--}0.020\text{ m}^2/\text{s}^2$. In the 2nd mixing stage, within distance of 1.35–1.78 m, the turbulent kinetic energy reaches $0.010\text{--}0.020\text{ m}^2/\text{s}^2$. As the mixing distance increases, the turbulent kinetic energy gradually decreases in each mixing stage, exhibiting the characteristic of the turbulent kinetic energy at the center of the chamber being greater than that at the periphery.

Regions with higher turbulent kinetic energy typically exhibit stronger hydraulic vortices and greater energy consumption. The intensity of turbulent kinetic energy for different mixers from highest to lowest is observed as COCM, Kenics, and HOCM. This indicates that COCM generates the largest hydraulic vortices and energy consumption, followed by the Kenics mixer, with HOCM exhibiting the lowest intensity. The concentrated vortex regions in the Kenics mixer are mainly distributed at the intersections of the mixing blades at each stage, suggesting that alternating flow reversals lead to a significant increase in

energy consumption. The hydraulic vortex regions of HOCM and COCM are primarily located at the diversion and collection groove, indicating that these components have a significant impact on the mixing process. Additionally, the closed-cavity structure of COCM generates vortices and energy consumption significantly higher than the hollow-cavity structure of HOCM.

3.2.3. The pattern of changes in the volume fraction distribution of the coagulant solution

The distribution of coagulant solution volume fraction at different cross-sections of each mixer was calculated using the Mixture Model represented by Eqs. (1)–(4) (Fig. 6). For Kenics mixer, the coagulant solution is diverted to the right chamber due to the obstruction of the blades in the 1st mixing stage, then the coagulant solution is split into two counter-rotating vortices by blades with opposite twisting directions, and it undergoes a rotational mixing motion on both sides of the blades in the 2nd mixing stage (Fig. 6a). In the subsequent third, fourth, and fifth mixing stages, the twisting directions of the blades alternate, continuously dividing the coagulant solution during the flow direction changes. Mixing gradually intensifies during this process, reaching a predominantly uniform mixing effect in the fifth mixing stage

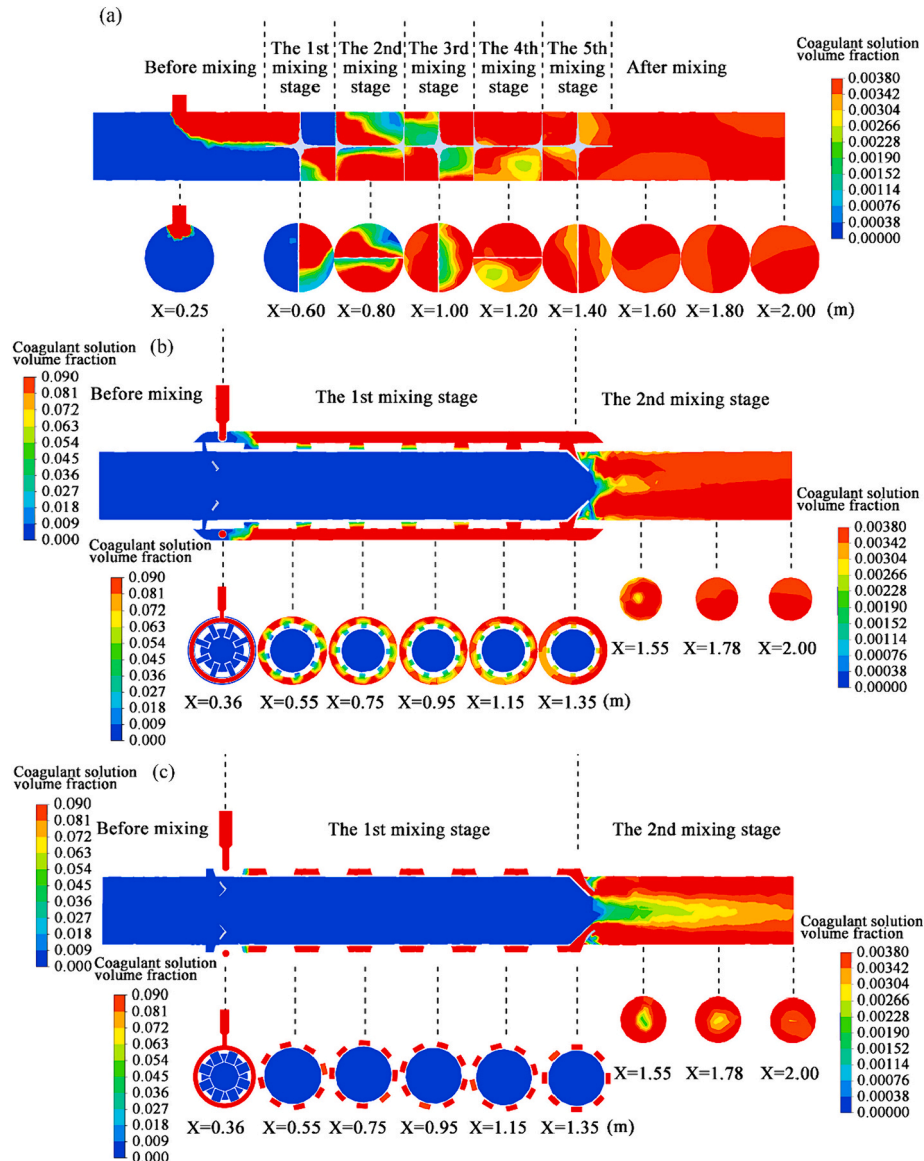


Fig. 6. The distribution of coagulant solution volume fraction at different cross-sections of Kenics (a), HOCM (b) and COCM (c).

(Fig. 6a). This mixing process agrees with the three-dimensional chaotic flow process in the Kenics mixer reported by Hobbs et al. [10].

In the mixing process in the HOCM, the distribution of coagulant solution volume fraction shows a helical flow trend as the twisting form of the guide groove changes in the 1st mixing stage, which results in a relatively uniform distribution of the coagulant solution in the mixing outer cavity with lower volume fractions inside the guide groove and higher volume fractions in the outer cavity, and achieves a relatively uniform mixing degree at the water collection groove (Fig. 6b). In the 2nd mixing stage, the inflow from the inner cavity is mixed with the inflow from the outer cavity, leading to rapid diffusion of the coagulant solution across the entire cross-section, and then a uniform mixing effect is achieved in the pipe section after 1.55 m.

The volume fraction of coagulant solution in each guide groove is relatively high in the 1st mixing stage of the COCM, and there is no significant difference in coagulant solution concentration at various cross-sections (Fig. 6c). During mixing in the 2nd mixing stage, the coagulant solution volume distribution is characterized by a lower concentration in the center of the pipe section and a higher concentration in the outer wall region, and tends to reach a uniformly mixed state with the increasing of mixing distance, and finally achieves a uniform mixing effect at the outlet of the mixer. In the 2nd mixing stage, the shortest distance required for HOCM to achieve uniform mixing is lower than that of COCM, which is caused by the different proportions of coagulant solution and inflow in the 1st mixing stage. In the 1st mixing stage, the mean volume fraction of the coagulant solution is 0.05 (HOCM with hollow outer cavity design) and 0.11 (COCM with closed outer cavity design), respectively, which means that more inflow flows into the outer cavity to mix with the coagulant solution in the HOCM mixer, resulting in higher dilution ratio and lower volume fraction of the coagulant solution in the first-stage mixing stage. Peng et al. demonstrated that the pre-dilution process has a positive effect on improving mixing efficiency [16]. Therefore, a higher dilution factor in the 1st mixing stage of HOCM helps to enhance the mixing efficiency of the 2nd mixing stage and shorten the mixing distance.

3.2.4. The variation pattern of mixing efficiency

There is no difference in the CoV values at the same cross-section of the Kenics mixer under various flow conditions (Fig. 7a). The mean CoV values exhibit an increasing trend before mixing, and reached the maximum value (1.480) at $X = 0.6$ m. In the mixing stages, the mean CoV values show a gradual decline with a decreasing rate at each stage. In the five-stage mixing process, the mean CoV values are 0.575, 0.274, 0.167, 0.110, and 0.061, respectively, and the mean CoV value at the outlet cross-section further decreases to 0.041, meeting the standard of complete mixing uniformity. It indicates that the 1st and 2nd mixing stages are the primary mixing zones of the Kenics mixer, and the 3rd to 5th mixing stages have a limited impact on improving the mixing efficiency. The result is consistent with the conclusion that the reduction in CoV values became significantly less pronounced as the number of mixing stages increased for Kenics mixer [45]. Increasing the number of mixing stages will inevitably lead to the increase of head losses and energy loss of the Kenics mixer, therefore, the HOCM and COCM mixers are designed based on the principle of reducing the number of mixing stages and achieving the swirl mixing effects, and their mixing efficiency are investigated.

There is no significant difference in CoV value of the same cross-section of the HOCM under different flow conditions, and stable mixing can be achieved within the range of 0.2–1.8 m/s (Fig. 7b). The mean value of CoV reaches the maximum value (1.177) at $X = 0.55$ m, and decreases to 0.856 at $X = 1.35$ m after the 1st mixing stage (the mean CoV decreased by 28.18 %). In the 2nd mixing stage, the mean CoV decreases rapidly between $X = 1.35$ m and $X = 1.55$ m (the mean CoV decreased by 71.72 %), and continues to decrease to 0.042 at the outlet cross-section, reaching the standard of complete uniformity. The results indicate that the primary mixing zone of the HOCM is the 2nd mixing stage, where the dilution of the 1st mixing stage and the resulting swirling flow promote the rapid reduction of CoV value.

There is no significant difference in CoV value of the same cross-section of the COCM under different flow conditions (Fig. 7c). The mean value of CoV reaches a maximum value of 0.790 at $X = 0.36$ m, and then decreases rapidly in the 1st mixing stage, dropping to 0.098 at $X = 0.55$ m, and maintains a low value throughout the 1st mixing stage

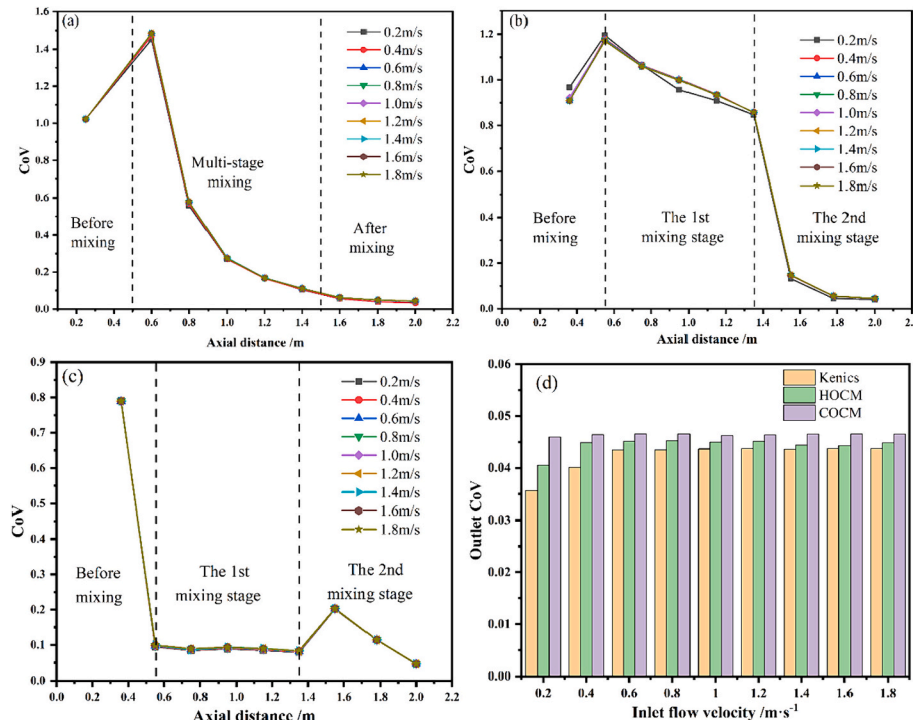


Fig. 7. The CoV variations along the Kenics mixer (a), HOCM (b), COCM (c), and the outlet cross-section (d).

(the mean value of CoV decreased by 95.16 %). At the 2nd mixing stage, the mean CoV briefly increases to 0.203 at $X = 1.35$ m to 1.55 m due to the mixing of water from the inner cavity and the coagulant solution from the outer cavity, and then rapidly decreases due to continuous swirl mixing (the mean CoV decreased by 4.84 %), and drops to 0.046 at the outlet section. It indicates that the 1st mixing stage is the primary mixing zone for the COCM mixer, where effective mixing is achieved as the coagulant solution enters the guide grooves. However, it is worth noting that there is a transient decrease in mixing efficiency during the confluence of the 2nd mixing stage. Therefore, it is essential to ensure an adequate length of the straight pipe section in the 2nd mixing stage to guarantee a thorough mixing process.

Under the flow rate of 0.2–1.8 m/s, the CoV values at the outlet cross-sections of three mixers can be maintained below 0.05, which can be considered as a complete mixing state (Fig. 7d). The CoV value of each mixer is Kenics mixer, HOCM, and COCM in order from low to high,

indicating that the Kenics mixer exhibits superior mixing performance compared to the HOCM and COCM, especially at low flow velocities. However, both HOCM and COCM mixers have lower head loss and G-values than Kenics mixer, and both meet the criteria for complete mixing, indicating that they are significantly superior to Kenics mixers in reducing fluid resistance and energy consumption, achieving a more efficient mixing process. Moreover, the CoV value, head loss and G-value of the HOCM are all smaller than those of the COCM under various flow conditions. Therefore, HOCM can be used as the optimal design scheme due to its better mixing effect and economy.

3.3. Exploration of hybrid mechanisms

To better illustrate the mixing development process from the microscopic level, the diagram of velocity vector and streamlines inside three mixers at a flow velocity of 1.0 m/s were plotted to observe the

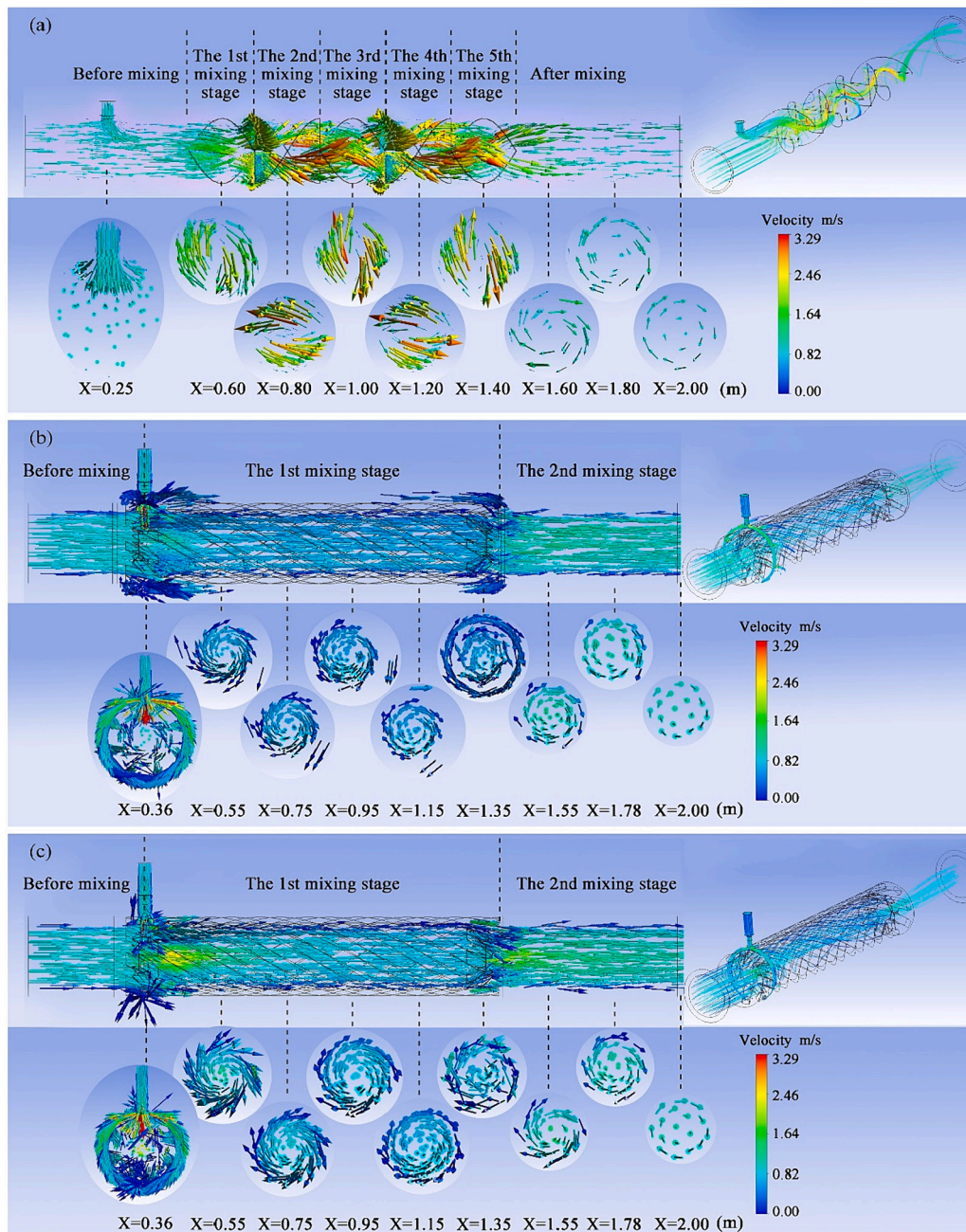


Fig. 8. Velocity vector diagrams and streamlines of Kenics (a), HOCM (b) and COCM (c) at a flow velocity of 1.0 m/s in flow patterns during the mixing process.

changes in flow patterns during the mixing process (Fig. 8). The velocity magnitude and direction of the mixing fluid undergo significant changes when it passes through each mixing stage of Kenics mixer attribution to the different twisting directions of the mixing blades at each stage (Fig. 8a). The direction of the flow line changes continuously along the mixing blades, exhibiting a characteristic alternating clockwise and counterclockwise, in which the fluid close to the blade surfaces is subjected to stronger resistance and forced steering, and the G-value increases significantly, resulting in a vortex phenomenon [46]. According to Kolmogorov theory, the presence of inertial forces and viscous resistance leads to velocity differences in the outer edges of large-scale vortices, which causes shear forces to be generated on the vortices and leads to their gradual splitting into smaller-sized vortices with greater shear forces and velocity gradients [47]. During this process, mixing efficiency is enhanced. However, the strong hydraulic vortices are generated due to the obstructive effect of the mixing blades, which converts the fluid kinetic energy into thermal energy and dissipates with the vortices, resulting in energy consumption and a significant increase in head loss.

The mixing mechanism of HOCM and COCM is quite different from that of Kenics (Fig. 8b and c). The inner and outer cavities of HOCM and COCM mixers produce continuous clockwise helical flows during the mixing process. In comparison to Kenics mixers, the flow pattern changes in HOCM and COCM effectively reduce the drastic variations in flow velocity during the mixing process and greatly enhance the flow capacity of mixers, which explains the lower head loss and G-values observed in Fig. 4 compared to Kenics mixers. From the velocity vector diagrams at various cross-sections, the velocity of HOCM and COCM mixers is higher at the central axis and mainly points in the direction of the axis (outlet), and lower near the pipe wall and points to the tangent direction of the pipe radius, which leads to gradient changes in the local velocity vectors during the swirling process and promotes the mixing process. Additionally, at the diversion ($X = 0.36$ m) and convergence ($X = 1.35$ m) cross-sections, the velocity vector becomes more disordered and the flow lines inside the mixer cross each other, resulting in a large velocity gradient and enhancing the mixing process, which explains the reason why CoV values drop sharply after $X = 0.36$ m and $X = 1.35$ m in Fig. 7b and c.

3.4. Comparison of mixing efficiency among different mixers

The hydraulic conditions, actual dimensions, and structural parameters of mixers significantly affect the head loss and mixing efficiency during the mixing process, making it challenging to directly compare the mixing effectiveness of mixers under different operating conditions. The head loss and mixing degree of HOCM and COCM are compared with those of traditional mixers of different types but the same size. The local friction factor and maximum mixing efficiency for different types of mixers are referenced from the summary by Valdes et al. [8], and the method for calculating head loss is described in Eq. (13), with a mixing length-to-diameter ratio (L/D) of 5 ($L = 1$ m, $D = 0.2$ m), and a fluid density of 1000 kg/m^3 . The head loss and maximum mixing efficiency of each type of mixer under different flow velocities are presented in Table 1.

Table 1
Variations in head loss and maximum mixing efficiency across different mixers.

Head loss for different types of mixers/m	0.2/ $\text{m}\cdot\text{s}^{-1}$	0.4/ $\text{m}\cdot\text{s}^{-1}$	0.6/ $\text{m}\cdot\text{s}^{-1}$	0.8/ $\text{m}\cdot\text{s}^{-1}$	1/ $\text{m}\cdot\text{s}^{-1}$	1.2/ $\text{m}\cdot\text{s}^{-1}$	1.4/ $\text{m}\cdot\text{s}^{-1}$	1.6/ $\text{m}\cdot\text{s}^{-1}$	1.8/ $\text{m}\cdot\text{s}^{-1}$	Maximum mixing efficiency/%
Kenics	0.05	0.21	0.46	0.82	1.29	1.85	2.52	3.29	4.16	96
SMX	0.17	0.69	1.54	2.74	4.29	6.17	8.4	10.97	13.89	99
SMV	0.07	0.27	0.62	1.1	1.71	2.47	3.36	4.39	5.56	96
ISG	0.1	0.41	0.93	1.65	2.57	3.7	5.04	6.58	8.33	98
Inliner Lightnin	0.06	0.25	0.56	0.99	1.54	2.22	3.02	3.95	5	96
HOCM	0.01	0.02	0.05	0.09	0.14	0.2	0.27	0.35	0.45	95
COCM	0.03	0.1	0.22	0.4	0.62	0.88	1.2	1.56	1.96	95

It can be observed that among traditional static mixers, such as Kenics, SMV, and Inliner Lightnin, the head loss ranges from 0.05 m to 5.56 m under flow velocities of 0.2–1.8 m/s, and achieving a maximum mixing degree of 96 %. For mixers like SMX and ISG, the head loss ranges from 0.1 m to 13.89 m, demonstrating they can achieve maximum mixing degrees of 98 %–99 % with higher head loss. The head loss for HOCM and COCM ranges from 0.01 m to 1.96 m, which are significantly lower than traditional mixers, and they can achieve a maximum mixing degree of 95 %.

This indicates that HOCM and COCM can greatly reduce the power consumption and energy loss in actual water treatment plant mixing processes. They can also reduce fluctuations and vortices, contributing to maintaining water stability, reducing mixer wear, and improving water treatment efficiency, energy consumption, and maintenance costs. The structural optimization methods for further reducing head loss and improving mixing efficiency still hold significant research value.

Future studies could focus on investigating the comprehensive effects of parameters such as the number of spiral diversion troughs, twist angles, lengths of diversion and collection groove, inner and outer cavity diameter ratios, etc., on mixing efficiency. Additionally, validating the mixing effectiveness of actual-sized mixers can help HOCM and COCM achieve optimal mixing degrees under low mixing resistance conditions.

4. Conclusion

This study employed a multiphase flow Mixture Model based on the Reynolds-averaged Navier-Stokes equations to investigate the mixing process of a novel multi-stage static pipeline mixer. The mathematical model exhibited high accuracy and can reliably predict head loss, G-values and mixing efficiency by comparing experimental and simulated mixing processes of 3D-HOCM and 3D-COCM mixers. The average relative errors between simulated and experimental values of head loss were ± 4.86 % (3D-HOCM) and ± 2.61 % (3D-COCM). The average relative errors between simulated and experimental values of G-values were ± 2.52 % (3D-HOCM) and ± 1.33 % (3D-COCM). Experimental mixing performance ($\text{CoV} < 0.02$) was superior to the simulated mixing performance ($\text{CoV} < 0.05$) for both mixers.

Compared to the Kenics mixer, HOCM and COCM can achieve efficient mixing with low energy consumption based on numerical simulations of the mixing process attribution to co-rotational vortex. The head loss of HOCM and COCM mixers was significantly reduced compared to the Kenics mixer, decreasing by 89.2 % (HOCM) and 52.6 % (COCM). The G-values were also significantly lower than the Kenics mixer, decreasing by 67.2 % (HOCM) and 31.1 % (COCM), respectively. The CoV values at the outlet cross-sections of three mixers can be maintained below 0.05. Under flow velocities ranging from 0.2 to 1.8 m/s, the head loss of the HOCM mixer was significantly lower than that of various types of traditional mixers, and achieved a maximum mixing degree of 95 %. HOCM can be regarded as the optimal design solution for a new type of multi-stage static pipeline mixer.

CRediT authorship contribution statement

Jiaheng Zhao: Conceptualization, Formal analysis, Investigation,

Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xing Li:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision. **Zedong Lu:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **Wenjun Chi:** Formal analysis, Funding acquisition, Resources, Supervision. **Zhibin Wang:** Formal analysis, Funding acquisition, Resources, Supervision. **Yawei Chen:** Conceptualization, Formal analysis, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that might influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

The Supporting Information file for this work can be found in the e-version of this paper online.

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